GenieUltra Hybrid Hardware Blueprint (Technical Preview)

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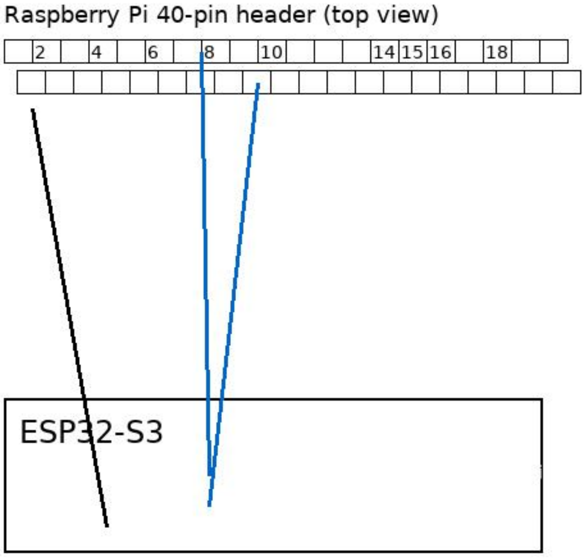
Legend:

Orange: HTTP / WebSocket

Blue: UART (JSON frames, 115200 8N1)

Green: BLE Notify / Advertise

Pinout & Wiring - Raspberry Pi 4 ↔ ESP32-S3 DevKitC



Protocol Flow & Frame Examples

Sequence: 1) Web UI (PWA) connects to Raspberry Pi hub (WebSocket) using the web-token segment. 2) Pi sends JSON frames over UART to ESP32 at 115200 8N1. Frames must include the sync token. 3) ESP32 parses JSON, performs BLE actions, and responds with status frames or BLE notifications. Sample command (Pi -> ESP32): {"cmd":"radio_play","station":2,"token":"<sync-token>"} Sample response (ESP32 -> Pi): {"status":"playing","station":2,"rssi":-52} UART settings: 115200 baud, 8 data bits, no parity, 1 stop bit (8N1). Use newline-terminated JSON for framing.

Power System: 5V Solar + LiPo Battery (Technical)

Recommended power architecture (safe, reliable for Pi + ESP32): 1) Solar Panel: 6V-12V nominal, UV-rated panel (e.g., 6V 10W) mounted outdoors. 2) Solar Charge Controller: use a board designed for LiPo solar charging (MCP73871-based or Adafruit Solar LiPo Charger with power-path management). It handles MPPT / power-path and charging from solar input. 3) Battery: 1-cell LiPo 3.7V nominal, capacity 5000-10000 mAh depending on expected runtime. 4) Power Path: Solar -> Charge Controller -> LiPo battery. The controller provides regulated output to a DC-DC boost converter. 5) Boost Converter: high-efficiency 5V boost capable of 3A (e.g., Pololu 5V 3A step-up or USB power bank module). This supplies Raspberry Pi 5V rail. Key protections & notes: - Use a low-voltage cutoff in charge controller or battery management (do not discharge LiPo below ~3.0V). - Include a Schottky blocking diode between panel and controller if recommended by module. - Add a fuse (2A slow-blow) and TVS transient protector on 5V output for surge protection. - For reliability, sized battery should provide at least 6-12 hours of typical usage. Wiring summary: Solar Panel (+) -> Solar IN of Charge Controller Solar Panel (-) -> Charge Controller GND Charge Controller BAT+ -> LiPo + Charge Controller BAT- -> LiPo - Charge Controller OUT -> Boost Converter VIN Boost Converter 5V OUT -> Raspberry Pi 5V pin (Pin 2) and VIN of ESP32 (optional) Common GND across all modules.

BOM, Enclosure & Deployment Notes

Bill of Materials (approximate): - Raspberry Pi 4 Model B (4GB) .... $55 - ESP32-S3 DevKitC .... $10 - 6V 10W UV-rated Solar Panel .... $18 - LiPo Battery 3.7V 5000 mAh .... $12 - Solar LiPo Charge Controller (MCP73871)....... $18 - Boost Converter 5V 3A .... $8 - MicroSD 8GB .... $5 - Case, mounting, jumper wires .... $6 Enclosure & Thermal: - Use a vented IP65 outdoor-rated enclosure for the solar-facing components. - Keep Pi in a ventilated compartment with a heat sink; consider a small fan if ambient >40°C. Deployment Checklist: 1) Flash Raspberry Pi OS and enable UART & SSH. 2) Install Python deps and deploy ble_radio_proxy_linked.py. 3) Flash ESP32 firmware and test UART JSON exchange. 4) Bench-test solar charging and measure runtime. Next steps: finalize PCB for internal wiring, add battery level telemetry to Web UI, and create safety labeling.